TEXORA · SESSIONAL NOTES · EXPERIMENT SHEET

House Wiring, Layout and the Single Line Diagram of a Factory

From the service pole to the last light point, an installation is a chain of protection and distribution. The house plan shows where every point goes; the factory's single line diagram shows how bulk power is tamed, metered and shared out.

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1. Aim and theory

Aim: to study the wiring of a house — service connection, meter, main switch and distribution board, sub-circuits, switches, fan and light points and earthing — to draw the wiring layout of a small house, and to draw and read the single line diagram (SLD) of a factory's electrical distribution.

House wiring. Power reaches a house through the **service connection**: a drop cable from the utility pole to the building, entering through a service pipe to the **energy meter**, then the consumer's **main switch / MCB distribution board (DB)**. From the DB the load is split into **sub-circuits** — by the usual wiring rules a light-and-fan sub-circuit carries at most about 800 W and 10 points, and a power sub-circuit about 2,000–3,000 W — each protected by its own fuse or MCB, so a fault trips one sub-circuit, not the house. Live conductors are always the ones switched; the neutral is never fused or switched alone. Every point is wired in parallel across the 230 V supply. Two safety systems complete the installation: **earthing** — a low-resistance path (earth electrode, earth wire run to every socket's earth pin and every metal body) that keeps exposed metal at earth potential and gives fault current a path heavy enough to blow the protection — and an **ELCB/RCCB**, which trips within milliseconds when it senses current leaking to earth through, for example, a person. Wiring systems in houses are surface conduit, concealed conduit, or batten (TRS) wiring; concealed PVC conduit is the modern default.

Single line diagram. A factory takes power at high voltage (commonly 11 kV in Bangladesh) and steps it down in its own substation. The SLD draws the whole three-phase system as one line: incoming HT line, lightning arrester, isolator and HT breaker, the transformer, the LT panel with its air circuit breaker (ACB), busbar, metering CTs, a capacitor bank for power-factor correction, and the outgoing feeders through MCCBs to sub-distribution boards and finally to motor and lighting loads. One line carries all the information an engineer needs — ratings, protection, and the order of distribution — which is why the SLD is the first document consulted for operation, fault-finding and expansion.

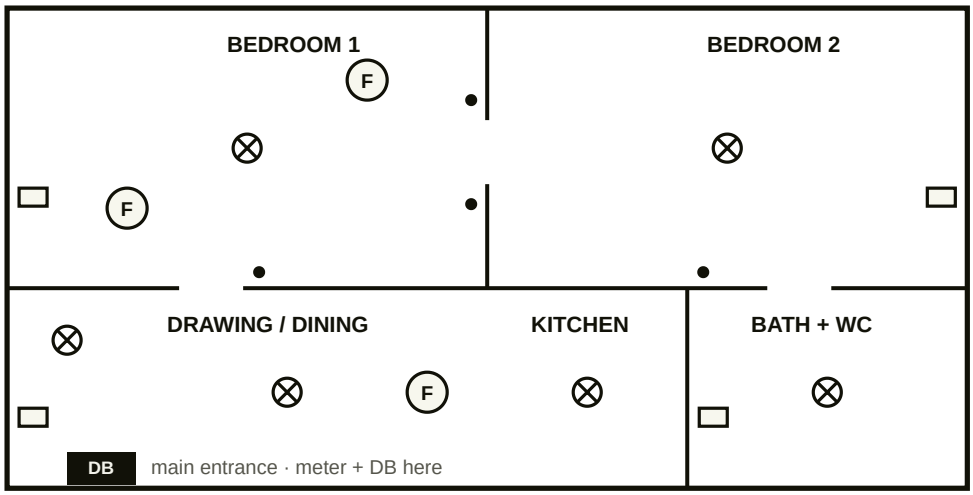


Figure 1 — Wiring layout of a small two-bedroom house. Every room gets a light point; fans in the bedrooms and drawing room; sockets on the walls; a switch board beside each door; the meter and distribution board at the entrance, where the service cable arrives. White gaps in the walls are the doorways.

2. Apparatus and procedure

- **Apparatus (study):** a model house-wiring board or the laboratory's wired demonstration panel — service cable, energy meter, main switch, MCB distribution board, ELCB/RCCB, light, fan and socket points with their switches, earth wire run and earth electrode; megger (insulation tester) and test lamp.
- Trace the path of power on the panel from the service entry to each point, naming every protective device in order and stating what each one protects against.
- Identify the live, neutral and earth conductors by their colours and confirm that every switch and fuse sits in the live conductor only.
- Draw the wiring layout of the small house in Figure 1, marking light, fan, socket and switch points room by room, and group the points into sub-circuits at the DB (light/fan sub-circuits separate from power sub-circuits).
- Test the demonstration installation: operate the ELCB test button, measure the earth-continuity and insulation resistance with the megger (power off), and record acceptable values (insulation resistance at least 1 MΩ; earth resistance ideally below 5 Ω, the lower the better).
- Draw the factory SLD of Figure 2 from the equipment list of a typical substation, labelling every item with its rating and function.

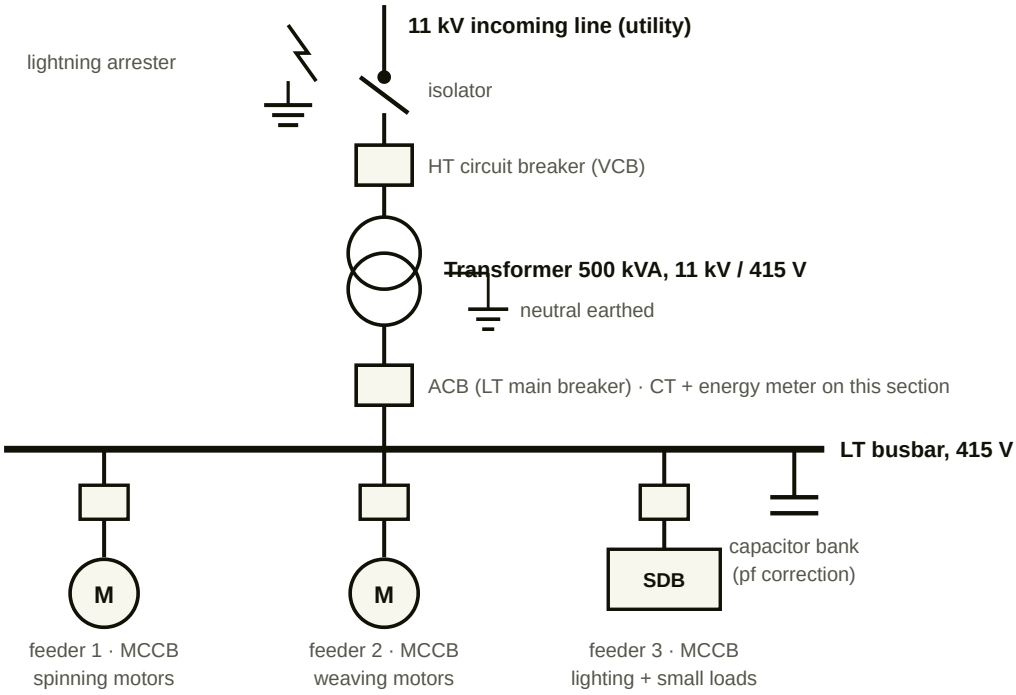


Figure 2 — Single line diagram of a small textile factory's substation and LT distribution: 11 kV in, protection and metering, one 500 kVA transformer, a 415 V busbar, motor and lighting feeders, and a capacitor bank holding the power factor up.

3. Worked calculation

Estimate the load of the house in Figure 1 and choose the main protection and cable. Assumed loads: 6 LED light points × 9 W = 54 W; 3 ceiling fans × 75 W = 225 W; 4 socket outlets allowed at 150 W each = 600 W; refrigerator 150 W; one air conditioner 1,500 W.

$$\text{Total connected load} = 54 + 225 + 600 + 150 + 1{,}500 = 2{,}529 \text{ W}$$

$$\text{Full-load current} = 2{,}529 / 230 = 11.0 \text{ A}$$

Choose the next standard MCB rating above the load with working margin: a **16 A main MCB**, with a 2.5 mm² copper service cable (current rating ≈ 20 A) so the cable is protected by the MCB, not the other way round. Split at the DB: **light/fan sub-circuit** — 54 + 225 = 279 W, a 6 A MCB on 1.5 mm² cable; **power sub-circuit** — 600 + 150 + 1,500 = 2,250 W ≈ 9.8 A, a 16 A MCB on 2.5 mm² cable. Note how diversity (not everything runs at once) makes the estimate comfortable in practice, but the wiring is sized for the full connected load regardless.

4. Precautions

- Never work on a live installation: isolate at the main switch, lock or tag it, and prove dead with a test lamp before touching a conductor.

- Fuses and MCBs belong in the live conductor only; a fuse in the neutral can leave a "dead" appliance live to earth after it blows.
- Earth continuity is part of the circuit, not an extra: a socket without its earth wire, or an earth wire joined with a loose twist, is a shock waiting for a fault. The earth pin of a three-pin socket must never be left unconnected or looped to neutral.
- Match cable size to the protective device: the MCB must trip before the cable overheats, so the cable's rating always exceeds the MCB's.
- Keep the meter and the utility's seals untouched — everything up to the meter is the supply authority's property; the consumer's responsibility starts at the main switch.

Practice (viva preparation)

1. List, in order, the equipment power passes through from the service pole to a light point in a house.
2. Why are loads divided into sub-circuits, and what are the usual limits of a light/fan sub-circuit?
3. What is the difference between an MCB and an ELCB/RCCB in what each detects?
4. In the factory SLD, what does the capacitor bank do, and why is it connected at the LT busbar?
5. A 3,450 W load is to run from a 230 V supply. Find the current and choose a suitable MCB rating.

Answer key

1. Service drop cable → service pipe/entry → energy meter → main switch → MCB distribution board → sub-circuit MCB → switch → light point (with the earth wire run alongside throughout).
2. So that a fault or overload trips only its own sub-circuit, fault-finding is easy, and cable sizes stay economical. Usual limits: about 800 W and 10 points per light/fan sub-circuit.
3. An MCB detects overcurrent — overload and short circuit — in the conductors it carries. An ELCB/RCCB compares live and neutral currents and trips on the small imbalance (typically 30 mA) that means current is leaking to earth, possibly through a person.
4. It supplies the lagging reactive power the induction motors would otherwise draw from the supply, raising the power factor toward unity — cutting line current, losses and the utility's low-pf penalty. It sits at the busbar so one bank corrects the whole plant's load at the point where all feeders share the supply.
5. I = 3,450/230 = 15 A. Choose a 20 A MCB (next standard size above 15 A) with cable rated above 20 A.
